SEP490 · CAPSTONE PROJECT · REVIEW 2 · BM01

Design Brief – Review 2

0. Header

Project code	**FA26SE032**
Project name	Sales Collaborator and Affiliate Network Management System (SCANMS)
Team	Nguyễn Thành Thắng (Leader - SE184251), Nguyễn Đình Tuấn (SE182540), Nguyễn Phú Quý (SE180104), Phan Xuân Thịnh (SE184527)
Supervisor	ThS. Tôn Thất Hoàng Minh
Repository link	https://github.com/thang-nt25/SCANMS-project-capstone-fall26 (tag: review-2)
Version	2.1 (Post-Review 2 Realigned Enterprise In-House Scope)
Date	08/11/2026

1. Scope

1.1 Registration Changes

Section	Current	Proposed	Reason
Timeline & Team (Sec. 1–2)	09/2026–03/2027; 5th unassigned row	07/09/2026–30/12/2026; finalized 4 members	Match Fall 2026 defense calendar and lock team roster.
Domain Model & Pricing (Sec. 3.2a–b)	Open external multi-store portal; unguarded discounts	4 in-house brand divisions, Category Lock, EAN-13 check & MAP floor-price promotions	Focus strictly on enterprise collaborator management, not an open C2C marketplace.
5-Role IAM & SoD (Sec. 3.2b–c)	3 roles; public shop signup; guest checkout	5 roles; Admin-only staff provisioning; KOC eKYC (Bronze 10%); 0-loss auth checkout	Prevent external shop creation and ensure verified tax/delivery identity.
Remuneration & Attribution (Sec. 3.2b)	Simple/additive rates; generic vouchers	Best-Benefit Precedence (cap 25%), KOC coupon [KOC][5|10], Self-Referral = 0%, KPI escrow	Protect brand margins, ensure cookie-less attribution, and block self-buying.
Returns, Escrow & Payout (Sec. 3.2b–c)	Linear payout; no return split; manual file only	14-day hold; 1-1 Exchange (keeps KOC fee) vs Refund (clawback); 10% PIT & BIDV Sandbox	Fairly protect KOCs on carrier damage while preventing overpayment on refunds.
Realtime, AI, Cloud & Mobile (Sec. 3.2b,f)	Static web portal; no chat/SLA/AI/mobile	Socket.IO VIP deals (>16% Admin gate), 14d Sample SLA, Dual AI, Cloud Web + Expo Mobile App	Support real-time partner workflows across deployed Web and Android clients.

1.2 Out of Scope

Item	Reason
Open External Store Marketplace & Public Shop Registration	SCANMS governs external collaborators for in-house enterprise brand divisions; external external third parties cannot open public stores.
Generic Retail Voucher Stacking & Standalone Marketing Portals	Platform/freeship voucher stacking and standalone promotional portals are excluded to prevent margin cannibalization; only MAP-guarded ProductPricePromotion and KOC-bound attribution coupons are supported.
Factory Manufacturing, Supplier Procurement (ERP/AP) & Robotics WMS	Upstream OEM manufacturing, purchase orders, and warehouse hardware belong to corporate ERP; SCANMS manages sellable stock quotas and SKU floor prices.
Proprietary Courier Vehicles & Licensed Core Banking Rails	Physical parcel delivery and fiat banking rails are delegated to GHN Sandbox API, PayOS (order collection), and BIDV Sandbox API (collaborator payouts).
Standalone Live Video Streaming Platform & Core Media Production	Livestream is strictly an auxiliary creator channel with product pinning (+2% bonus capped at 25%); SCANMS does not provide an RTMP video encoding farm or open streaming platform.

2. System at a Glance

SCANMS solves fragmented partner onboarding, attribution fraud, margin erosion, and manual commission settlement for multi-brand D2C retail enterprises managing external sales collaborators (KOLs/KOCs). Built for five roles—System Admin, System Manager, Shop Manager, Collaborator, and Customer—across deployed Web portals (React/Vite) and a native Android application (React Native/Expo), SCANMS automates the entire affiliate lifecycle across four company-owned brand divisions. Core capabilities include Admin-governed staff provisioning, Maker-Checker eKYC, category-locked catalogs, MAP floor-price promotions, 4-axis AI partner matchmaking, Socket.IO VIP deal negotiation with >16% Admin budget escalation, 14-day sample review SLAs, Smart Link/QR and KOC-coupon attribution, 14-day post-delivery Adaptive Escrow ledger matching customer return warranty with 1-for-1 exchange vs. refund clawback, and row-locked payouts with 10% PIT withholding via BIDV Sandbox. See System Overview Architecture Diagram in SDD (Report 4, Section 3.1).

3. Key Design Decisions

ID	Decision	Options considered	Chosen option and rationale	Owner	Related FR / diagram
D1	In-House Brand Provisioning & Shared Mobile/Web Bootstrap	Public shop self-signup & separate mobile BFF vs Admin-only staff provisioning & unified NestJS API	Blocked public SHOP_MANAGER signup (/admin/personnel only) and exposed /api/mobile/bootstrap over one NestJS modular monolith so Web and Expo Android enforce identical 5-role RBAC.	Nguyễn Thành Thắng	F-01, F-02 / SDD Sec. 3.1
D2	MAP Floor-Price Guard & KOC-Exclusive Attribution Coupon	Unrestricted shop price edits & multi-voucher stacking vs Maker-Checker MAP guard & KOC-bound coupon	Enforced proposedPrice >= floorPrice with System Manager approval (ProductPricePromotion) and single KOC goods coupon ([KOC][5|10]) for deterministic cookie-less attribution.	Nguyễn Thành Thắng	F-06, F-13 / SDD Sec. 4.2
D3	Socket.IO VIP Deal Negotiation with >16% Admin Escalation	HTTP polling & unguarded deal rates vs WebSocket rich deal cards with 2-tier approval gate	Used Socket.IO /chat rooms with autonomous activation for 10%–16% deals and mandatory PENDING_ADMIN_APPROVAL escalation to System Admin for 17%–25% deals.	Phan Xuân Thịnh	F-09, F-10 / SDD Sec. 5.3
D4	14-Day Post-Delivery Hold & Dual-Path Return Resolution	Instant commission release or blanket clawback on all claims vs 14-day hold with Exchange vs Refund split	Held pending commissions 14 days post-delivery: 1-for-1 EXCHANGE on carrier damage preserves 100% KOC commission (+ GHN claim), whereas REFUND triggers automatic clawback.	Nguyễn Phú Quý	F-16, F-17 / SDD Sec. 5.4
D5	Best-Benefit Commission Precedence & Row-Locked BIDV Payout	Additive rate stacking & unlocked balance updates vs Precedence override, KPI escrow & SELECT FOR UPDATE	Applied min(25%, max(VIP Deal, 10%+Tier, 10%) + Live), isolated VIP bonus in conditionalDealEscrowBalance, and locked wallet rows with 10% flat PIT withholding via BIDV Sandbox.	Nguyễn Đình Tuấn	F-17, F-19 / SDD Sec. 6.2

4. Top Risks

ID	Risk	Impact	How we verify or mitigate	Owner
R1	Self-Referral Purchases & Bot Click Flooding distorting attribution and draining marketing budgets.	High: Unfair commission leakage and skewed AI conversion rates (CR%).	isSelfReferral guard forces self-purchase commission to 0%; Redis sliding-window limit (100 req/min/IP) with RAM LRU fallback and AiFraudService anomaly flags.	Nguyễn Phú Quý

ID	Risk	Impact	How we verify or mitigate	Owner
R2	Brand Margin Erosion from below-cost discounts or excessive VIP collaborator rates.	High: Negative unit economics on discounted SKUs.	assertFloor(proposedPrice, floorPrice) Maker-Checker approval, Best-Benefit non-additive cap (<=25%), >16% Admin approval gate, and KPI-locked deal bonus escrow.	Nguyễn Thành Thắng
R3	Concurrent Withdrawal Double-Spending, Unreturned Samples, or Refund Overpayment.	High: Wallet overdrafts, sample inventory loss, and unrecovered commissions.	PostgreSQL SELECT ... FOR UPDATE wallet locks, idempotent BIDV Sandbox inquiry, 14-day return hold clawback, and 14-day sample SLA overdue quota lock.	Nguyễn Đình Tuấn

5. Known Gaps

ID	Gap	Location	Plan	Owner	Due
G1	SDD ERD & Architecture Docs (21-table draft) lag behind 69-model Prisma Schema & Expo Mobile client (BM02 SR-01, SR-02)	docs/architecture/SAD_DOCUMENT.md, DATABASE_DESIGN.md vs schema.prisma & Mobile/	Regenerate Logical ERD, State Machines, and System Overview diagram in Report 4 to match 69 Prisma models, Vite Web, Expo Mobile, and @nestjs/schedule.	Nguyễn Thành Thắng	Defense 1.1
G2	Mobile Android App defers Customer PayOS Checkout & Live Room to Web Portal in current iteration (BM02 SR-04)	Mobile/src/features/cart/CartScreen.tsx, Mobile/docs/ROLE_WORKFLOWS_2026-11-03.md	5-role Mobile operations (catalog, KYC, chat, VIP deals, samples, pricing, GHN orders, returns, payouts) are active; complete native checkout deep-link handoff & staging APK build.	Phan Xuân Thịnh	Defense 1.1
G3	Commercial Host-to-Host Bank Disbursement simulated via BIDV Sandbox Adapter (BM02 SR-05)	backend/src/modules/payouts/bidv-payout.adapter.ts	Kept by design due to banking license constraints; BIDV Sandbox idempotency, inquiry reconciliation (SBX-...), FinancePosting, and VietQR batch export verified.	Nguyễn Đình Tuấn	Kept (Sandbox)
G4	Residual deprecated database/mobile stubs after scope removal of standalone promotional portals (BM02 SR-03)	backend/prisma/schema.prisma, Mobile/src/features/workspace/policy.ts	Web routes already redirect/exclude legacy flows and rate parameter is fixed to 0; hide remaining mobile menu entry and mark legacy DB table deprecated in SDD.	Nguyễn Thành Thắng	Defense 1.1

6. Proposed Mandatory Features

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-01	Multi-Role Auth, Registration/Reset Email OTP & Argon2id Security	Collaborator, Customer	Argon2id (>=64MB RAM) hashes passwords; 6-digit Email OTP verifies new registrations & password resets (>=5 wrong OTPs invalidate code); Login alerts via email; unauthorized Web routes or /api/mobile/bootstrap calls return 403 Forbidden.	Unit, E2E	**DONE**
F-02	Enterprise 5-Role RBAC & Privilege Escalation Guard	System	Strict RBAC enforces separation of duties across SYSTEM_ADMIN, SYSTEM_MANAGER, SHOP_MANAGER, COLLABORATOR, and	Integration, UI	**DONE**

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
			CUSTOMER; public registration payload with role=SHOP_MANAGER is coerced to CUSTOMER.		
F-03	Internal Brand Staff & In-House Store Provisioning	System Admin	Only SYSTEM_ADMIN can create SHOP_MANAGER (bound to storeId & GHN satellite warehouse) and SYSTEM_MANAGER accounts; blocks public shop creation and logs immutable audit trail.	Integration, UI	**DONE**
F-04	Collaborator eKYC Dossier & Maker-Checker Verification [BR-01]	Collaborator, System Manager	Collaborator submits 12-digit CCCD (front/back photos), bank account & TikTok/FB/YouTube URL; System Manager independent audit verifies CCCD matches bank account name 100%; approval transitions KycStatus=VERIFIED, activates Bronze Tier (10% base rate), and assigns collaborator to in-house Store.	Integration, UI	**DONE**
F-05	2-Tier Product Variant Matrix Management & EAN-13 Barcodes	Shop Manager	Shop Manager creates 2-tier variants (Color/Size, Scent/Volume); validates unique EAN-13 barcodes across inventory; stock decrements atomically across warehouse reserves.	Integration, UI	**DONE**
F-06	MAP Floor-Price Guard & Product Price Moderation [BR-06]	Shop Manager, System Manager	System Manager configures SKU floorPrice; single/batch price promotions with proposedPrice < floorPrice return 400 Bad Request; discount activates only when status=ACTIVE within budget.	Integration, UI	**DONE**
F-07	4-Axis AI Partner Matchmaking & Radar Scoring Engine	Shop Manager, Collaborator	AI engine scores verified Collaborators (0–100 across Category Relevance, Conversion Rate, Social Tier, Price Fit) in <=500ms; bilateral partnership invite transitions PENDING -> APPROVED.	Integration, UI	**DONE**
F-08	Socket.IO Realtime Deal Chat & Media/Asset Sharing	Shop Manager, Collaborator	Real-time bi-directional chat allowing Shop Managers and Collaborators to negotiate terms and directly exchange product photos, review guidelines, and promotional assets with end-to-end integrity.	Integration, UI	**DONE**
F-09	Autonomous Deal Negotiation (<=16%) & Admin Escalation (17–25%) [BR-08]	Shop Manager, System Admin	Counterparties exchange rich VIP Deal cards over Socket.IO chat; 10%–16% deals activate immediately on accept; accepted 17%–25% deals escalate to System Admin queue; proposals >25% fail with 400.	Integration, UI	**DONE**
F-10	0 VND Sample Product Sponsorship & 14-Day Video Review SLA [BR-10]	Collaborator, Shop Manager	Verified Collaborators request 0 VND product samples (capped at maximum 3 active borrowed samples per store); Shop dispatches sample via GHN at 0 VND; receipt (RECEIVED) triggers 14-day SLA countdown.	Integration, UI	**DONE**
F-11	Sample Video Review Verification & Overdue Borrowing Lock	Shop Manager, Collaborator	Collaborator submits review video URL within 14 days for Shop verification (COMPLETED); overdue samples past 14 days transition to OVERDUE and lock new sample requests.	Integration, UI	**DONE**
F-12	Multi-Channel Attribution Asset Generation: Smart Links, QR & Coupons	Collaborator	Collaborator generates short referral link (/r/:shortCode), dynamic QR code, and exclusive deal goods coupon ([KOC][RATE], e.g., THAONHI16); public vouchers excluded.	Integration, UI	**DONE**
F-13	Authenticated Checkout, Anti-Self-Referral Guard [BR-12] & PayOS VietQR	Customer, System	Checkout requires login while preserving 100% cart items; if Customer identity matches Collaborator by userId, phone, or bank account, commission is forced to 0%; encrypted PayOS VietQR webhook marks order PAID instantly.	Integration, UI	**DONE**
F-14	Multi-Warehouse GHN Logistics Routing & Live Status Tracking [BR-13]	Shop Manager, GHN Courier	Shop Manager confirms order and dispatches GHN parcel from designated brand satellite warehouse (weight/dimensions); SKU stock decrements atomically and status tracks SHIPPING -> DELIVERED.	Integration, UI	**DONE**
F-15	14-Day Return Claim Submission with	Customer	Customer initiates return request within 14 days of delivery; damaged goods claim	Integration, UI	**DONE**

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
	Mandatory Unboxing Video [BR-14]		requires photographic proof and mandatory unboxing video to isolate transit fault.		
F-16	1-to-1 Damaged Item Exchange: 100% Commission & KPI Preserved [BR-14a]	Shop Manager, Customer	For transit damage (LOGISTICS_FAULT), Shop dispatches 1-to-1 replacement item at 0 VND; preserves 100% Collaborator commission and KPI progress; logs separate GHN compensation claim.	Integration, UI	**DONE**
F-17	Automated Return Refund & Clawback Execution [BR-14b]	System, System Manager	Approved return refunds credit customer internal shopping wallet; simultaneously executes automated Clawback reversing 100% of order's held commissions to 0 VND and deducts refunded GMV from Deal KPI.	Integration, UI	**DONE**
F-18	System Manager Platform Return Dispute Arbitration	System Manager	System Manager reviews escalated return disputes, inspects unboxing video and GHN tracking logs, and issues binding arbitration ruling (APPROVE_REFUND, APPROVE_EXCHANGE, UPHOLD_SHOP) protecting customers and collaborators.	Integration, UI	**DONE**
F-19	Stage 1: Auto-Release Standard Remuneration (10%+Tier) after 14D [BR-17]	System	Automated reconciliation engine matures 10% base commission + Tier bonus (10%–13% total) from PENDING_14D_HOLD to availableBalance exactly 14 days after successful delivery without dispute.	Integration, UI	**DONE**
F-20	Stage 2: Exclusive Deal KPI Settlement (Escrow Release vs Forfeit) [BR-18]	System, Shop Manager	Holds VIP bonus in conditionalDealEscrowBalance until Deal hits 100% KPI (QUALIFIED) or expires without KPI (FORFEITED, keeping standard commission). Closed-month FR-09 milestone settles cleanly.	Integration, UI	**DONE**
F-21	Row-Locked Collaborator Withdrawal (SELECT FOR UPDATE) & 10% Flat PIT [BR-19,20]	Collaborator, System Manager	SHOP_MANAGER withdrawal returns 403; COLLABORATOR request (>=200k VND) executes SELECT FOR UPDATE row-level lock on wallet; automatically withholds flat 10% personal income tax (FinancePosting).	Integration, UI	**DONE**
F-22	Disburse Napas 24/7 Sandbox & Executive Financial Ledger Reconciliation	System Admin, System	Disburses 90% net payout via Napas 24/7 Sandbox; automatically restores 100% funds on bank timeout (PAYOUT_REJECT_REFUND); updates real-time Executive Financial Ledger with zero reconciliation discrepancy.	Unit, E2E	**DONE**
F-23	Immutable Security & Financial Audit Trail	System	100% of staff creations, KYC decisions, MAP updates, VIP deal approvals, return resolutions & payouts append an AuditLog entry; API blocks edit/delete.	Unit, E2E	**DONE**
F-24	Category-Locked In-House Catalog & Vertical Isolation Guard	System	Backend service validates SKU creation against owning Store's locked primaryCategory (Cosmetics, Apparel, Nutrition, Smart Living); rejects cross-vertical listings.	Integration, UI	**DONE**
F-25	Sample Sponsorship Quota Guard & Borrowing Limits	System	Background service enforces store-level sample inventory quotas and restricts Collaborators to a maximum of 3 concurrent active borrowed samples across in-house stores.	Integration, UI	**DONE**
F-26	30-Day Attribution Session & Fraud-Resilient Click Tracking [BR-11]	System	Referral link click sets 30-day AttributionSession cookie with Last-Click attribution precedence; Redis rate-limiting restricts spam clicks at 100 req/min/IP.	Integration, UI	**DONE**
F-27	GHN Delivery Webhook Listener & Automated 14-Day Clock [BR-14]	System	Asynchronous webhook receiver processes GHN carrier status updates, transitions delivered shipments to DELIVERED atomically, and initializes the 14-day warranty return timer.	Integration, UI	**DONE**
F-28	Monthly Milestone Bonus Settlement Engine (FR-09 Auxiliary)	System	Scheduled end-of-month cronjob aggregates clean GMV across 14-day matured orders, evaluates tier milestone rules, and credits milestone bonuses to eligible Collaborator wallets.	Integration, UI	**DONE**

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-29	Automated KPI Forfeiture Resolution Engine	System	Automated scheduler evaluates expired Exclusive Deals, forfeits conditional deal escrow balances for unfulfilled targets, and returns escrow reservations to store budget.	Integration, UI	**DONE**
F-30	Executive Financial Ledger & Multi-Store Audit Reconciliation	System Admin, System	Enterprise financial balance sheet aggregating gross inflows, 14-day escrow liabilities, net payout disbursements, and 10% PIT tax obligations across all 4 in-house brand stores.	Unit, E2E	**DONE**

7. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method
NFR-01	API & Redirect Latency	REST & Smart Link p95 Response Time	<= 200 ms for catalog queries and /r/:shortCode redirection	Autocannon benchmark (500 concurrent requests) & NestJS timing interceptor
NFR-02	Authentication & RBAC Security	Password Hash Cost & Role Isolation	Argon2id (>= 64 MB memory, 3 iterations); 100% cross-role calls blocked (403)	Jest security suites (auth-otp.spec.ts, rbac guards) across Web & Mobile endpoints
NFR-03	Web & Mobile Client Responsiveness	Web Initial Bundle / FCP & Mobile Bootstrap	Web gzipped entry <= 500 KB, FCP <= 1.5 s; Android /mobile/bootstrap <= 300 ms	Vite build bundle audit, Lighthouse CI, and Mobile contract/startup test suite
NFR-04	Transactional & Inventory Integrity	Atomic Checkout & Return Rollback Rate	100% atomic rollback on stock exhaustion or invalid return state transition	Jest integration specs (orders.service.spec.ts, return-pickup.postgres.integration.spec.ts)
NFR-05	Rate Limiting & Cache Resilience	Click Throttle & Redis Outage Failover	100 req/min/IP limit; 0 unhandled 500 errors during simulated Redis disconnect	CacheService LRU memory fallback unit tests & click-flood load script
NFR-06	Financial Concurrency & Auditability	Double-Spend Prevention & Ledger Coverage	0 negative wallet balances under 20 parallel payouts; 100% mutations in AuditLog	Parallel SELECT ... FOR UPDATE payout concurrency test & FinancePosting assertions

8. Ownership Map

Member	Role	Owned parts (2–3)	Main workflow owned	Key decisions / PRs
Nguyễn Thành Thắng (Leader)	Fullstack Architect & Platform Governance Lead	WF-01: F-01..F-04; WF-05: F-19..F-22; Platform: F-23, F-28, F-30 (11 features total)	WF-01: Staff Provisioning & KOC eKYC; WF-05: Two-Stage Escrow & 10% PIT Payouts	Decisions D1 & D5; System Governance, IAM, Ledger & Executive Dashboard
Phan Xuân Thịnh	Fullstack Developer & Catalog/Realtime Lead	WF-02: F-05..F-09; Platform: F-24, F-29 (7 features total)	WF-02: Moderate Products, AI Match & Negotiate Exclusive Deals	Decision D2; Products, Variants, Floor Price & Chat Negotiation

Member	Role	Owned parts (2–3)	Main workflow owned	Key decisions / PRs
Nguyễn Đình Tuấn	Fullstack Developer & Attribution Lead	WF-03: F-10..F-13; Platform: F-25, F-26 (6 features total)	WF-03: Fulfill Experience Samples, Issue Marketing Tools & Capture Authenticated Attribution Orders	Decision D3; Marketing Toolkit, Attribution Engine & Fraud Sentinel
Nguyễn Phú Quý	Fullstack Developer & Logistics/Arbitration Lead	WF-04: F-14..F-18; Platform: F-27 (6 features total)	WF-03: Experience Samples, Marketing Tools & Attribution Orders	Decision D4; GHN Logistics, 1-to-1 Exchange vs Refund Clawback

Appendix A – Defense 1.1 Targets

A1. Main Workflows

ID	Workflow	Actors	Trigger	Business outcome	Features	Entity states	Owner
WF-01	WF-01: Provision Internal Brand Staff & Onboard Verified Collaborators	System Admin, Customer, System Manager, Collaborator	System Admin appoints an internal Shop Manager (/admin/personnel), or a Customer registers, verifies Email OTP, and submits a KOC upgrade eKYC dossier (/customer/upgrade).	In-house Shop Manager bound to assigned brand store; verified Customer upgraded to Collaborator at Bronze (10%) tier with verified social channel and immutable AuditLog.	**F-01, F-02, F-03, F-04**	User: isActive=true (SHOP_MANAGER/COLLABORATOR); KycStatus: UNVERIFIED -> VERIFIED / REJECTED; Tier: Bronze (10%)	Nguyễn Thành Thắng
WF-02	WF-02: Moderate Product Variants, AI Match Collaborators & Negotiate Exclusive Deals	Shop Manager, System Manager, Collaborator	Shop Manager creates a 6-block product SKU, proposes a price drop, uploads Socket.IO Deal Chat & Media assets, and runs 4-Axis AI Partner Matchmaking.	Category-locked SKU moderated to APPROVED, MAP-compliant price promotion activated, Top KOC matched & approved for store cooperation, and HD media downloaded.	**F-05, F-06, F-07, F-08, F-09**	ProductModerationStatus: DRAFT -> APPROVED / REJECTED; PricePromotionStatus: PENDING -> APPROVED -> ACTIVE; StoreCollaboratorStatus: PENDING -> APPROVED	Phan Xuân Thịnh
WF-03	WF-03: Fulfill Experience Samples, Issue Marketing Tools & Capture Authenticated Attribution Orders	Collaborator, Shop Manager, GHN Courier, Customer	Collaborator requests 0 VND product sample, generates Smart Link, QR & Coupon; Customer places authenticated order.	Sample dispatched under 14-day SLA; creator generates attribution tools; zero-loss checkout succeeds with Anti-Self-Referral verification.	**F-10, F-11, F-12, F-13**	SampleRequestStatus: PENDING -> APPROVED -> SHIPPED -> RECEIVED -> VIDEO_SUBMITTED -> COMPLETED / OVERDUE; AttributionSession: ACTIVE; OrderStatus: PENDING -> PAID	Nguyễn Đình Tuấn
WF-04	WF-04: Fulfill Multi-Warehouse Logistics, Resolve 14-Day Returns & Execute Clawback Reversal	Collaborator, Customer, Shop Manager, System Manager	Collaborator shares Smart Link/QR and KOC Attribution Coupon; Customer checks out, receives GHN parcel, and files either a damaged-item Exchange or a Refund return.	Order attributed accurately (0% if self-referral), shipped via brand GHN warehouse; 1-for-1 EXCHANGE preserves 100% KOC commission (+ GHN claim) while REFUND claws back commission.	**F-14, F-15, F-16, F-17, F-18**	OrderStatus: PENDING -> SHIPPING -> DELIVERED -> COMPLETED / RETURNED; ReturnRequestStatus: REQUESTED -> SHOP_APPROVED -> INSPECTING -> EXCHANGE_SHIPPED / REFUNDED -> COMPLETED; CommissionStatus: PENDING -> APPROVED / REVERSED	Nguyễn Phú Quý

ID	Workflow	Actors	Trigger	Business outcome	Features	Entity states	Owner
WF-05	WF-05: Settle Two-Stage Commission, Withhold 10% PIT & Disburse Payouts	Shop Manager, Collaborator, System Manager, System Admin	14-day post-delivery hold elapses on eligible orders, Shop Manager settles closed-month FR-09 milestones, and Collaborator submits a bank withdrawal request.	Mature commissions and qualified KPI bonuses unlock to availableBalance, monthly bonus & tier upgrade apply once, 10% flat PIT is withheld under DB row lock, and net payout disburses via BIDV Sandbox.	**F-19, F-20, F-21, F-22**	CommissionStatus: PENDING -> APPROVED; DealBonusStatus: PENDING_KPI -> QUALIFIED / FORFEITED; PayoutStatus: PENDING -> PROCESSING -> APPROVED / REJECTED	Nguyễn Thành Thắng

A2. Demo Scenarios

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01a	WF-01	Main	Seeded System Admin, System Manager, and new Customer account; Sora Skincare store configured.	1. Admin creates internal SHOP_MANAGER bound to Sora Skincare at /admin/personnel. 2. Customer registers, verifies OTP, and submits CCCD, MB Bank & TikTok channel at /customer/upgrade. 3. System Manager approves eKYC on Web/Mobile.	Staff account active on Sora Skincare; Customer upgraded to COLLABORATOR with KycStatus=VERIFIED, initial tier Bronze (10%), and AuditLog recorded.	Nguyễn Thành Thắng
DS-01b	WF-01	Exception	Unauthenticated client sends role=SHOP_MANAGER to /auth/register; KOC submits blurred CCCD or >5 wrong OTPs.	1. Call POST /auth/register with role=SHOP_MANAGER. 2. Enter wrong OTP 5 times. 3. System Manager rejects mismatched KYC bank/CCCD dossier with mandatory reason.	Public signup coerced to CUSTOMER (privilege escalation blocked); OTP invalidated after 5 failures; KycStatus=REJECTED with feedback and resubmission enabled.	Nguyễn Thành Thắng
DS-02a	WF-02	Main	Sora Skincare (locked to Cosmetics), verified KOC profiles, and SKU floorPrice = 350,000 VND.	1. Shop Manager creates Vitamin C Serum (450k VND, EAN-13); System Manager approves. 2. Shop Manager proposes 390k VND promotion (>= floorPrice); System Manager approves. 3. Run 4-Axis AI Matchmaker, approve KOC cooperation, and KOC downloads HD media.	Product APPROVED on storefront; promotion ACTIVE at 390k VND; AI returns ranked KOC radar scores; StoreCollaborator=APPROVED and HD asset downloaded.	Phan Xuân Thịnh
DS-02b	WF-02	Exception	Shop Manager attempts cross-category SKU, duplicate EAN-13 barcode, or price drop below floorPrice (320k < 350k VND).	1. Submit Electronics category from Cosmetics store. 2. Submit duplicate EAN-13 barcode. 3. Propose promotion price 320,000 VND below 350,000 VND MAP floor.	All three requests rejected with 400 Bad Request; product and price remain unchanged, protecting brand vertical isolation and unit margin.	Phan Xuân Thịnh

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-03a	WF-03	Main	Approved Sora–KOC partnership on Web/Mobile; approved Vitamin C Serum with sample quota > 0.	1. KOC requests 0 VND sample for Vitamin C Serum -> Shop dispatches GHN at 0 VND. 2. KOC submits review video URL within 14 days -> Shop approves COMPLETED. 3. KOC generates Smart Link (/r/:shortCode), Dynamic QR & Coupon (THAONHI16). 4. Customer signs in, preserves cart, applies coupon, and pays via PayOS VietQR.	Sample completed within 14-day SLA; zero-loss checkout succeeds with 10% coupon discount and attribution logged to KOC.	Nguyễn Đình Tuấn
DS-03b	WF-03	Exception	VIP Deal proposed at 28% (>25% cap); KOC misses 14-day sample review deadline; pinned SKU stock hits 0.	1. KOC misses 14-day review deadline and attempts new sample request. 2. Collaborator attempts self-referral purchase using own smart link with matching userId/phone/bank. 3. Client spams referral link exceeding 100 req/min.	Overdue sample transitions to OVERDUE and blocks new requests; Anti-Self-Referral Guard [BR-12] forces commission to 0%; Redis rate-limiter returns 429.	Nguyễn Đình Tuấn
DS-04a	WF-04	Main	Verified KOC with active 16% VIP deal & KOC coupon (10% off); authenticated Customer; Sora GHN warehouse.	1. KOC generates /r/:shortCode, QR & KOC coupon. 2. Customer clicks link / applies KOC coupon, logs in preserving 100% cart, pays via PayOS/COD. 3. Shop creates GHN waybill -> DELIVERED. 4. On transit damage, Customer chooses EXCHANGE -> Shop ships 1-1 replacement & opens GHN claim.	Order attributed to KOC with pending commission recorded; replacement delivered (COMPLETED), 100% KOC commission preserved, and GHN compensation claim opened.	Nguyễn Phú Quý
DS-04b	WF-04	Exception	KOC attempts self-purchase via own link/coupon; Customer requests full return & refund within 14-day window.	1. KOC checks out using own referral link/matching phone. 2. Separate Customer files REFUND return with unboxing video -> parcel returned & inspected -> refund approved.	Self-referral order sets commission rate = 0% and flags AiFraudService; refunded order credits Customer shoppingBalance and reverses KOC commission to REVERSED.	Nguyễn Phú Quý

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-05a	WF-05	Main	Delivered order past 14-day hold with no open return; closed-month clean GMV > 50M VND; verified KOC wallet.	1. Reconciliation job matures 14-day PENDING commission to availableBalance and qualifies KPI deal escrow. 2. Shop settles closed-month FR-09 milestone bonus & promotes KOC tier. 3. KOC requests 2,072,000 VND withdrawal -> System Manager approves via BIDV Sandbox.	Wallet debited once under SELECT ... FOR UPDATE; 207,200 VND (10% PIT) posted via FinancePosting; 1,864,800 VND net approved with SBX-... reference and AuditLog.	Nguyễn Thành Thắng
DS-05b	WF-05	Exception	SHOP_MANAGER attempts withdrawal; KOC requests < 200,000 VND or fires parallel requests exceeding balance; BIDV Sandbox timeout.	1. Call POST /payouts/withdraw as SHOP_MANAGER. 2. Submit 100,000 VND (<200k min) and 2 concurrent full-balance requests. 3. Simulate BIDV timeout/rejection.	SHOP_MANAGER blocked (403 Forbidden); <200k rejected (400); row lock allows at most 1 request (0 negative balance); rejected payout restores 100% funds via PAYOUT_REJECT_REFUND.	Nguyễn Thành Thắng

A3. Functional Test Case Target

Source	Count	Minimum per item	Subtotal
Mandatory features (from Section 6)	30	3	90
Main workflows (end-to-end, from A1)	5	1	5
State transitions of main entities (KYC, Product, PricePromo, Deal, Sample, Live, Order, Return, Commission, Payout)	15	1	15
Agreed NFRs & concurrency/security suites (from Section 7)	4	1	4
Committed total (minimum)	—	—	**114**